

From Primitives to Metamaterials and Photonics

A Walkthrough of the Event Density Derivation

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1 The Question

1.1 What this walkthrough derives

This walkthrough derives, from substrate primitives, the structural backbone of four Nobel-frontier results in metamaterials and photonics:

1. **The effective constitutive tensors** $\varepsilon_{\text{eff}}^{ij}(\omega)$ and $\mu_{\text{eff}}^{ij}(\omega)$ of a periodic subwavelength microstructure, derived via a two-scale homogenization expansion and a cell problem on a unit cell Y . The cell-problem solution defines a corrector field $\chi^j(y)$ whose cell average provides the substrate-level correction to the naive arithmetic average of the microstructure response.
2. **The negative refractive index** $n_{\text{eff}}(\omega) = -\sqrt{\varepsilon_r(\omega)\mu_r(\omega)}$ when both effective tensors are negative in an overlapping frequency band. The negative branch is FORCED by causal outgoing-wave selection (Pendry 2000). The Drude form $\varepsilon_{\text{eff}}(\omega) = \varepsilon_{\infty}(1 - \omega_p^2/\omega^2)$ from wire-array microstructures and the Lorentz form $\mu_{\text{eff}}(\omega) = 1 - F\omega^2/(\omega^2 - \omega_0^2 + i\gamma\omega)$ from split-ring resonators are each derived as cell-problem outputs.
3. **The constitutive transformation under substrate deformation** $\varepsilon'^{ij} = (\det J)^{-1}J^i_k J^j_l \varepsilon^{kl}$ and the coarse-grained **effective metric** $g_{ij} = (J^{-1})^k_i (J^{-1})^l_j \delta_{kl}$ on the wave equation, with the wave equation rewritten in covariant Laplace-Beltrami form. The metric is a coarse-grained rule-type metric on the substrate, not spacetime curvature.
4. **The Pendry spherical cloak** (Pendry 2006). The radial map $r' = R_1 + \alpha r$ with $\alpha = (R_2 - R_1)/R_2$ produces a shell-supported anisotropic medium whose effective metric vanishes radially at the inner shell. The cloaked interior is topologically excluded from the substrate seen by the propagating channel.
5. **The metasurface generalized Snell's law** (Capasso 2011). An engineered codimension-1 surface carrying a tangential phase profile $\Phi(r_{\parallel})$ imposes a tangential momentum kick $k_{\parallel,2} = k_{\parallel,1} + \nabla_{\parallel}\Phi$, yielding

$$n_1 \sin \theta_i = n_2 \sin \theta_t + \frac{\lambda}{2\pi} \frac{d\Phi}{dx}.$$

6. **The photonic bandgap** (Yablonovitch 1987) as a frequency window in which no Bloch eigenmode exists on the periodic substrate at $a \sim \lambda$. The relevant Bloch theorem is derived inline in compact form.
7. **The unifying structural identity:**

Metamaterials = engineered microstructure+engineered deformation+engineered discontinuity.

The three operations exhaust the substrate-level wave-control space within the coarse-graining window $\ell_P \ll a \ll \lambda \ll L$.

Each result is derived, not posited. Every required two-scale expansion step, cell-problem statement, Jacobian computation, effective-metric identification, jump condition, and Bloch-eigenmode statement appears inline.

1.2 Why standard photonics treats ε , μ , and metamaterial response phenomenologically

Conventional macroscopic electromagnetism treats ε , μ , and the refractive index n as phenomenological material parameters fit to experiment. The Maxwell equations are taken as exact at the macroscopic level, with material response encoded in constitutive relations $\mathbf{D} = \varepsilon \mathbf{E}$ and $\mathbf{B} = \mu \mathbf{H}$. The standard derivations of ε and μ from molecular polarizabilities (Clausius-Mossotti, Lorentz local-field corrections) treat the microscopic constituents as fixed atoms or molecules whose responses are summed over a coarse-graining volume.

The recognition that subwavelength structuring can engineer ε_{eff} and μ_{eff} unattainable in natural materials — including negative values — was Pendry’s central insight. The further recognition that smooth coordinate transformations of Maxwell’s equations correspond to anisotropic constitutive tensors that can be physically built by metamaterials was the transformation-optics insight. The realization that engineered codimension-1 phase profiles produce anomalous refraction outside the bulk Snell relation was the metasurface insight. The recognition that periodic dielectric structures at $a \sim \lambda$ produce band structures with frequency gaps was the photonic-bandgap insight.

Standard theory delivers the macroscopic equations correctly. It is silent about what ε and μ are ontologically: it treats them as numbers attached to a material, with no account of *why* a particular microstructure produces a particular response, *why* cloaking works mechanistically rather than merely formally as a coordinate identity, *what* a metasurface boundary condition is at the level of substrate ontology, or *why* the four Nobel-frontier directions share a common structural skeleton.

1.3 What Event Density claims

In the ED framework the substrate carries rule-type structure at scale ℓ_P , where rule-type means the discrete pattern of participation-rule alignment that the substrate locally enforces on channels of ED-flow. Coarse-graining within the window $\ell_P \ll a \ll \lambda \ll L$ produces an effective wave equation whose coefficients are coarse-grained projections of the substrate’s local rule-type content. Three independent engineered modulations of this content control wave propagation:

- **Smooth periodic microstructure** at scale $a \ll \lambda$ produces effective constitutive tensors via homogenization. The substrate is unstrained; only its internal rule-type content is modulated. Pendry-negative-index lives here.
- **Smooth deformation** $u(X)$ produces an effective metric and transformed constitutive tensors via transformation optics. The substrate is strained; its rule-type content is transported covariantly along the deformation. Pendry-cloak lives here.
- **Engineered codimension-1 discontinuity** with engineered phase profile produces tangential momentum kicks at the interface. The substrate is unstrained in the bulk; only the interface carries engineered rule-type content. Capasso-metasurface lives here.
- **Periodic microstructure** at $a \sim \lambda$, adjacent to the homogenization regime, supports Bloch eigenmodes with band structure. Yablonovitch-bandgap lives here.

The substrate-level statement, in one sentence: **metamaterials are engineered modulations of the substrate’s coarse-grained rule-type structure; the three independent modulation axes (microstructure, deformation, discontinuity) exhaust the wave-control space within the coarse-graining window.**

1.4 The chain in summary

The derivation chain runs: substrate primitives P-MM-1 through P-MM-6 (§2) $\rightarrow$ two-scale lift, derivative splitting, order-by-order equations, cell problem, corrector field, averaging operator,

effective-tensor formula (§3) → identification with ε and μ , Drude wire-array response, Lorentz split-ring response, doubly-negative band, FORCED negative-index branch (§4) → deformation Jacobian, rule-type deformation tensor, constitutive transformation, effective metric in covariant form, Pendry spherical cloak, topological exclusion of cloaked interior (§5) → codimension-1 discontinuity, phase-imprinting boundary condition, tangential momentum kick, generalized Snell’s law (§6) → Bloch form, band structure, bandgap formation, relation to homogenization (§7) → unifying substrate-level statement, three-axis exhaustiveness, scope limits (§8) → FORCED / INHERITED / OPEN accounting (§9) → exact claims (§10).

2 The Substrate Primitives

Six substrate primitives are used. No new primitives are introduced anywhere in the derivation; the entire walkthrough runs on this list.

2.1 P-MM-1. Substrate rule-type microstructure

The substrate carries rule-type structure at scale ℓ_P — discrete participation rules indexed by spatial position and internal type, encoding the substrate’s local commitment patterns for ED-flow. At coarse-grained scales $a \gg \ell_P$, engineered inclusions modulate this rule-type structure, producing a position-dependent coarse-grained response. The coefficients $\varepsilon^{ij}(x)$, $(\mu^{-1})^{ij}(x)$, and $n^2(x)$ of the macroscopic wave equation are coarse-grained projections of this rule-type content.

Substrate-level meaning. ε and μ are not fundamental fields. They are coarse-grained labels for the substrate’s local rule-type response: $\varepsilon^{ij}(x)$ is the coarse-grained polarizability of the rule-type structure at x — the local tendency to align ED-flow under an applied electric tension — and $(\mu^{-1})^{ij}(x)$ is the coarse-grained inverse circulation susceptibility — the local resistance to closed-loop ED-flow under an applied magnetic tension.

Engineered modulation. Replacing the substrate’s vacuum rule-type content with an engineered pattern of inclusions modulates these coefficients. The pattern of inclusions, coarse-grained, is the metamaterial response.

2.2 P-MM-2. Subwavelength periodicity

When inclusions are arranged with period a satisfying $a \ll \lambda$, the rule-type structure admits a two-scale description in a slow coordinate $X = x$ and a fast coordinate $y = x/a$, with y taking values on the unit cell $Y = [0, 1]^3$ (in normalized fast coordinates). Field quantities lift to two-scale form $\psi(x) \rightarrow \tilde{\psi}(X, y)$ with $\tilde{\psi}$ periodic in y . The two-scale expansion is the substrate-level mathematical reflection of the scale separation $a/\lambda \ll 1$.

Periodic substrate. The rule-type content is the same on each unit cell up to translations by aR with R in the integer lattice $\mathbb{Z}^3$. Practical metamaterials approximate this idealization to within a few percent at microwave frequencies.

2.3 P-MM-3. ED-gradient deformation

A slowly varying displacement field $u(X)$ acting on the substrate produces a coarse-grained deformation

$$x'^i = x^i + u^i(X),$$

whose Jacobian

$$J^i_j = \delta^i_j + \frac{\partial u^i}{\partial X^j}$$

transports rule-type structure covariantly. The deformation is smooth and orientation-preserving: $\det J > 0$ everywhere, J and J^{-1} bounded. Time-independent deformations carry no internal dispersion of their own; all dispersion enters through the constitutive coefficients at the microstructure level.

Substrate-level meaning. The deformation is a coordinate-level engineered displacement of the substrate. Rule-type tensors transform as densities of weight one: a coefficient $\varepsilon^{ij}(x)$ at the undeformed point x becomes the coefficient $\varepsilon'^{ij}(x')$ at the deformed point x' , with the transformation rule derived in §5.

2.4 P-MM-4. Channel propagation in structured media

Wave propagation in the coarse-grained substrate is governed by an effective second-order wave equation

$$\partial_{x^i} \left[A^{ij}(x) \partial_{x^j} \psi(x) \right] + k_0^2 n^2(x) \psi(x) = 0,$$

where ψ is a scalar component of the electromagnetic field (or any field reducible to scalar form for a chosen polarization), $A^{ij}(x)$ is a coarse-grained rank-2 tensor coefficient determined by the substrate's local rule-type content, $n^2(x)$ is a coarse-grained scalar coefficient, and $k_0 = \omega/c$ is the vacuum wavenumber.

Polarization identifications. For Maxwell's equations in the time-harmonic regime, the scalar form corresponds to:

- TM modes (transverse magnetic, $\mathbf{E}$ in the plane of incidence): $A^{ij} \leftrightarrow (\mu^{-1})^{ij}$, $n^2 \leftrightarrow \varepsilon$.
- TE modes (transverse electric, $\mathbf{H}$ in the plane of incidence): $A^{ij} \leftrightarrow \varepsilon^{ij}$, $n^2 \leftrightarrow \mu$.

The derivations of §3 apply to either polarization by reading the identifications.

2.5 P-MM-5. Interface rule-type discontinuity

A codimension-1 surface Σ across which the rule-type microstructure or deformation jumps imposes jump conditions on the coarse-grained wave field. In the absence of engineered surface phase profile or surface currents, the standard tangential-field continuity holds: $\mathbf{E}_\parallel$ and $\mathbf{H}_\parallel$ are continuous across Σ . An engineered surface phase profile $\Phi(r_\parallel)$ imposes a phase-jump matching condition derived in §6.

Substrate-level meaning. A metasurface is a codimension-1 engineered rule-type discontinuity: the substrate carries different rule-type content on the two sides of Σ , with the discontinuity itself engineered to imprint a controlled phase shift on transmitted channels.

2.6 P-MM-6. Coarse-graining window

All derivations operate in the window

$$\ell_P \ll a \ll \lambda \ll L,$$

where ℓ_P is the substrate cutoff (smallest substrate length scale), a is the microstructure period (engineered), λ is the operating wavelength, and L is the device scale.

Within this window:

- Substrate discreteness is invisible at the wavelength scale: $\ell_P \ll \lambda$ ensures that the coarse-grained wave equation is meaningful.
- Microstructure averaging is well-defined: $a \ll \lambda$ ensures that the two-scale expansion converges and the cell-problem corrector is well-posed.

- Device-scale envelopes are slow: $\lambda \ll L$ ensures that boundary effects at device boundaries do not invalidate bulk derivations.

Boundaries of the window. The Bloch / bandgap regime lives at $a \sim \lambda$, adjacent to but outside the strict homogenization regime $a \ll \lambda$. The substrate cutoff ℓ_P is invisible throughout this walkthrough; nothing here depends on its precise value.

3 Two-Scale Expansion and the Cell Problem

This section derives the homogenization framework that produces ε_{eff} and μ_{eff} . The two-scale expansion is the mathematical reflection of P-MM-2 and P-MM-6.

3.1 Setup

Under P-MM-1, P-MM-2, P-MM-4, P-MM-6, consider the scalar form of the wave equation

$$\partial_{x^i} \left[A^{ij}(x/a) \partial_{x^j} \psi(x) \right] + k_0^2 n^2(x/a) \psi(x) = 0,$$

where $A^{ij}(y)$ and $n^2(y)$ are Y -periodic functions of the fast coordinate $y = x/a$. The coefficient $A^{ij}(y)$ is positive-definite and bounded:

$$c_- |\xi|^2 \leq A^{ij}(y) \xi_i \xi_j \leq c_+ |\xi|^2, \quad 0 < c_- \leq c_+ < \infty,$$

for all $\xi \in \mathbb{R}^3$ and all $y \in Y$. The ratio a/λ is small; we expand the equation in this parameter.

3.2 Two-scale lift

Lift $\psi(x)$ to a two-scale function $\tilde{\psi}(X, y)$, treating $X = x$ (slow) and $y = x/a$ (fast) as independent variables. The chain rule gives

$$\frac{\partial}{\partial x^i} = \frac{\partial}{\partial X^i} + \frac{1}{a} \frac{\partial}{\partial y^i}.$$

This is the **derivative splitting** rule of two-scale analysis. It expresses the fact that a spatial gradient $\partial/\partial x$ has two contributions: a slow contribution from the device-scale envelope $\partial/\partial X$ and a fast contribution from the cell-scale modulation $a^{-1}\partial/\partial y$.

Lift ψ to an asymptotic series in a :

$$\tilde{\psi}(X, y) = \psi_0(X, y) + a\psi_1(X, y) + a^2\psi_2(X, y) + a^3\psi_3(X, y) + \dots,$$

with each $\psi_k(X, y)$ smooth in X and Y -periodic in y . Substitute into the wave equation and collect terms by power of a .

3.3 Order-by-order equations

The wave equation, in lifted form, reads

$$\left(\partial_{X^i} + a^{-1} \partial_{y^i} \right) \left[A^{ij}(y) \left(\partial_{X^j} + a^{-1} \partial_{y^j} \right) \tilde{\psi} \right] + k_0^2 n^2(y) \tilde{\psi} = 0.$$

Expand the bracket and collect powers of a .

Order a^{-2} :

$$\partial_{y^i} \left[A^{ij}(y) \partial_{y^j} \psi_0(X, y) \right] = 0.$$

Multiply by ψ_0 and integrate over Y ; integrate by parts using Y -periodicity (boundary terms cancel between opposite faces of Y):

$$\int_Y A^{ij}(y) \partial_{y^i} \psi_0 \partial_{y^j} \psi_0 d^3y = 0.$$

By positive-definiteness of A^{ij} (P-MM-4), the integrand is non-negative and vanishes only if $\partial_{y^j}\psi_0 = 0$. Therefore $\psi_0(X, y) = \psi_0(X)$ is independent of y . **The leading-order field has no microstructure dependence.**

Order a^{-1} :

$$\partial_{y^i} \left[A^{ij}(y) \partial_{y^j} \psi_1(X, y) \right] = -\partial_{y^i} \left[A^{ij}(y) \partial_{X^j} \psi_0(X) \right].$$

The right-hand side is linear in $\partial_{X^j}\psi_0$. By linearity of the elliptic equation in ψ_1 , the solution has the form

$$\psi_1(X, y) = \chi^j(y) \partial_{X^j} \psi_0(X) + \tilde{\psi}_1(X),$$

where the **corrector field** $\chi^j(y)$ satisfies the **cell problem**:

$$\boxed{\partial_{y^i} \left[A^{ij}(y) \partial_{y^j} \chi^k(y) \right] = -\partial_{y^i} A^{ik}(y), \quad \chi^k \text{ is } Y\text{-periodic, with } \langle \chi^k \rangle = 0.}$$

The cell problem is a linear elliptic boundary-value problem on Y with periodic boundary conditions. By the Fredholm alternative for elliptic operators with periodic boundary conditions, it has a unique solution up to additive constants; the zero-mean condition $\langle \chi^k \rangle = 0$ fixes the constant.

Substrate-level meaning of the corrector field. $\chi^k(y)$ is the local accommodation pattern of the substrate response to a unit slow gradient in the k -direction. The substrate's rule-type structure adjusts within each unit cell to absorb the slow gradient; χ^k records that adjustment.

3.4 The averaging operator

Define the cell-averaging operator

$$\langle f \rangle \equiv \frac{1}{|Y|} \int_Y f(y) d^3y.$$

Two basic properties follow from Y -periodicity:

1. **Vanishing of pure y -divergences.** For any Y -periodic vector field $V^i(y)$,

$$\langle \partial_{y^i} V^i \rangle = \frac{1}{|Y|} \int_Y \partial_{y^i} V^i d^3y = \frac{1}{|Y|} \oint_{\partial Y} V^i n_i dS = 0,$$

because the boundary integral on ∂Y vanishes by periodicity (opposite faces cancel).

2. **Commutativity with X -derivatives.** Since X and y are independent in the lifted picture,

$$\langle \partial_{X^i} f(X, y) \rangle = \partial_{X^i} \langle f(X, y) \rangle.$$

These two properties drive the homogenization computation.

3.5 Order- a^0 equation and effective tensor

Order a^0 :

$$\partial_{X^i} \left[A^{ij} \partial_{X^j} \psi_0 \right] + \partial_{X^i} \left[A^{ij} \partial_{y^j} \psi_1 \right] + \partial_{y^i} \left[A^{ij} \partial_{X^j} \psi_1 \right] + \partial_{y^i} \left[A^{ij} \partial_{y^j} \psi_2 \right] + k_0^2 n^2 \psi_0 = 0.$$

Apply the averaging operator. The pure $\partial_{y^i}[\dots]$ terms (third and fourth terms) vanish by property 1. The second term, using $\psi_1 = \chi^j \partial_{X^j} \psi_0$ and property 2, becomes $\partial_{X^i} \langle A^{ij} \partial_{y^j} \chi^k \rangle \partial_{X^k} \psi_0$. The first term becomes $\partial_{X^i} \langle A^{ij} \rangle \partial_{X^j} \psi_0$. The fifth term becomes $k_0^2 \langle n^2 \rangle \psi_0$.

Combining:

$$\partial_{X^i} \left[(\langle A^{ik} \rangle + \langle A^{ij} \partial_{y^j} \chi^k \rangle) \partial_{X^k} \psi_0 \right] + k_0^2 \langle n^2 \rangle \psi_0 = 0.$$

This is the **homogenized wave equation** at leading order:

$$\partial_{X^i} \left[A^{*ik} \partial_{X^k} \psi_0 \right] + k_0^2 n^{*2} \psi_0 = 0,$$

with the **effective coefficient**

$$A^{*ik} = \langle A^{ik} \rangle + \langle A^{ij} \partial_{y^j} \chi^k \rangle, \quad n^{*2} = \langle n^2 \rangle.$$

The effective coefficient is the cell-averaged coefficient *plus* a correction from the cell-problem solution. The correction captures the local accommodation of the field to the microstructure: as the field threads through the unit cell, it deforms to follow the substrate's rule-type response, and the deformation feeds back into the cell-averaged flux.

3.6 Voigt-Reuss bounds

The corrector correction is non-positive in the energy norm. Standard variational arguments on the cell problem give

$$\langle (A^{-1})^{ij} \rangle^{-1} \leq A^{*ij} \leq \langle A^{ij} \rangle$$

as quadratic forms on $\mathbb{R}^3$. The lower (Reuss) bound is the harmonic mean of A^{ij} over the cell; the upper (Voigt) bound is the arithmetic mean. The effective coefficient lives between these bounds and saturates the upper bound only when A^{ij} is constant on Y (no microstructure, trivial homogenization).

3.7 Worked example: layered substrate

For a one-dimensional layered substrate with $A(y) = A_1$ on layer 1 (volume fraction f) and $A(y) = A_2$ on layer 2 (volume fraction $1 - f$), the cell problem in 1D reduces to

$$\frac{d}{dy} \left[A(y) \left(\frac{d\chi}{dy} + 1 \right) \right] = 0,$$

so $A(y)(d\chi/dy + 1) = C$ (constant on Y). Solving:

$$\frac{d\chi}{dy} + 1 = \frac{C}{A(y)}, \quad \chi \text{ periodic} \Rightarrow \langle d\chi/dy \rangle = 0 \Rightarrow 1 = C \langle 1/A \rangle.$$

So $C = \langle 1/A \rangle^{-1}$, and the effective coefficient parallel to the layering is

$$A_{\parallel}^* = \langle A(d\chi/dy + 1) \rangle = \langle C \rangle = \langle 1/A \rangle^{-1},$$

the harmonic mean. Perpendicular to the layering, $A_{\perp}^* = \langle A \rangle$, the arithmetic mean. The microstructure thus produces anisotropy even from isotropic constituents — a substrate-level FORCED feature of any non-trivial periodic microstructure.

4 Effective Constitutive Tensors and Negative Index

4.1 From scalar form to Maxwell

For Maxwell's equations in a non-magnetic medium ($\mu = \mu_0$), the TE polarization wave equation in 2D is

$$\partial_i \left[\frac{1}{\varepsilon(x)/\varepsilon_0} \partial_i H_z \right] + \frac{\omega^2}{c^2} H_z = 0,$$

identifying $A^{ij} \leftrightarrow (\varepsilon/\varepsilon_0)^{-1} \delta^{ij}$. For TM polarization,

$$\partial_i \left[\frac{1}{\mu(x)/\mu_0} \partial_i E_z \right] + \frac{\omega^2}{c^2} E_z = 0,$$

identifying $A^{ij} \leftrightarrow (\mu/\mu_0)^{-1} \delta^{ij}$. Applying §3 in each polarization gives effective constitutive tensors.

4.2 Effective constitutive tensors

In full vector form, for an anisotropic periodic substrate with $\varepsilon^{ij}(y)$ and $\mu^{ij}(y)$ both Y -periodic, the homogenization theorem yields

$$\boxed{\varepsilon_{\text{eff}}^{ij} = \langle \varepsilon^{ij} \rangle + \langle \varepsilon^{ik} \partial_{y^k} \chi_E^j \rangle, \quad (\mu_{\text{eff}}^{-1})^{ij} = \langle (\mu^{-1})^{ij} \rangle + \langle (\mu^{-1})^{ik} \partial_{y^k} \chi_M^j \rangle,}$$

with $\chi_E^j(y)$ and $\chi_M^j(y)$ the electric and magnetic correctors solving their respective cell problems.

Anisotropy from microstructure. Even if $\varepsilon^{ij}(y) = \varepsilon(y)\delta^{ij}$ and $\mu^{ij}(y) = \mu(y)\delta^{ij}$ are locally isotropic, the corrector solutions χ_E^j , χ_M^j generally produce anisotropy in $\varepsilon_{\text{eff}}^{ij}$ and μ_{eff}^{ij} . The principal axes of the effective tensors are aligned with the symmetry axes of the microstructure.

4.3 Substrate-level meaning of ε and μ

In the ED picture, $\varepsilon^{ij}(y)$ is the substrate's local rule-type polarizability: the coarse-grained tendency for ED-flow to align under an applied electric tension. $(\mu^{-1})^{ij}(y)$ is the local rule-type inverse circulation susceptibility: the coarse-grained resistance to closed-loop ED-flow under an applied magnetic tension. Neither is a fundamental field; both are coarse-grained labels for rule-type response.

The corrector fields $\chi_E^j(y)$ and $\chi_M^j(y)$ are the local accommodation patterns of the substrate response. When a slow gradient $\partial_{X^j}\psi_0$ threads through a unit cell, the substrate's rule-type structure adjusts within the cell; $\chi^j(y)$ records the spatial pattern of that adjustment, and the cell-averaged adjustment contributes to the effective tensor.

4.4 Wire-array microstructure $\rightarrow$ Drude $\varepsilon_{\text{eff}}(\omega)$

Consider a square array of thin conducting wires of radius r_0 and lattice constant a , with $r_0 \ll a \ll \lambda$. Standard analysis (Pendry 1996) of the cell problem with electric field along the wire axis identifies the wires as supporting a longitudinal collective oscillation with effective plasma frequency

$$\omega_p^2 = \frac{2\pi c^2}{a^2 \ln(a/r_0)}.$$

The wire-array effective dielectric function takes the Drude form

$$\boxed{\varepsilon_{\text{eff}}(\omega) = \varepsilon_\infty \left(1 - \frac{\omega_p^2}{\omega^2} \right).}$$

Below ω_p , $\varepsilon_{\text{eff}}(\omega) < 0$. Above ω_p , $\varepsilon_{\text{eff}}(\omega) > 0$ and approaches ε_∞ at high frequency.

Substrate-level meaning. The wire array carries a collective rule-type oscillation: ED-flow along the wires is constrained by the wire geometry to a single longitudinal mode, with effective plasma frequency set by the geometric line density and the cell logarithm. Below this frequency, the substrate's longitudinal response is *opposite* to the applied tension — the engineered rule-type collective mode dominates over any local polarizability — producing negative effective ε .

4.5 Split-ring microstructure $\rightarrow$ Lorentz $\mu_{\text{eff}}(\omega)$

Consider an array of split-ring resonators (SRRs) — pairs of concentric metallic loops with a gap, embedded in a host dielectric — with lattice constant a , resonance frequency ω_0 , oscillator strength F (related to the ring's filling fraction), and damping rate γ . Standard analysis of the

cell problem for an applied magnetic field perpendicular to the rings (Pendry 1999) gives the Lorentz form

$$\mu_{\text{eff}}(\omega) = 1 - \frac{F\omega^2}{\omega^2 - \omega_0^2 + i\gamma\omega}.$$

For ω in a narrow band immediately above ω_0 , $\text{Re } \mu_{\text{eff}}(\omega) < 0$.

Substrate-level meaning. Each SRR is a closed-loop rule-type resonator: ED-flow circulating around the ring constitutes a coarse-grained magnetic moment, whose natural frequency ω_0 is set by the ring's geometric inductance and capacitance (gap capacitance). Driven above ω_0 , the rule-type circulation lags by more than $\pi/2$ and opposes the applied magnetic tension, producing negative effective μ .

4.6 The doubly-negative band

Engineering both wire arrays and SRRs into the same substrate produces a frequency band where both $\varepsilon_r(\omega) < 0$ and $\mu_r(\omega) < 0$. In this band, $\varepsilon_r\mu_r > 0$, so the wave equation

$$\nabla^2\psi + k_0^2\varepsilon_r\mu_r\psi = 0$$

admits propagating solutions $\psi = e^{i\mathbf{k}\cdot\mathbf{x}}$ with $|\mathbf{k}|^2 = k_0^2\varepsilon_r\mu_r > 0$.

4.7 Negative-index branch FORCED

The refractive index satisfies $n_{\text{eff}}^2 = \varepsilon_r\mu_r$, so $n_{\text{eff}} = \pm\sqrt{\varepsilon_r\mu_r}$. The branch is determined by causal outgoing-wave selection.

Argument. In the doubly-negative band, the relation between $\mathbf{D}$ and $\mathbf{E}$ is $\mathbf{D} = \varepsilon_r\varepsilon_0\mathbf{E}$ with $\varepsilon_r < 0$, reversing direction. Similarly, $\mathbf{B} = \mu_r\mu_0\mathbf{H}$ with $\mu_r < 0$ reverses direction. The Poynting vector

$$\mathbf{S} = \mathbf{E} \times \mathbf{H}$$

remains in the direction of energy outflow (causality, P-MM-4). For a plane wave, the phase velocity $\mathbf{v}_p = (\omega/|\mathbf{k}|^2)\mathbf{k}$ is parallel to $\mathbf{k}$. Combining Maxwell's curl equations $\nabla \times \mathbf{E} = -\partial_t\mathbf{B}$ and $\nabla \times \mathbf{H} = \partial_t\mathbf{D}$:

$$\mathbf{k} \times \mathbf{E} = \omega\mu_r\mu_0\mathbf{H}, \quad \mathbf{k} \times \mathbf{H} = -\omega\varepsilon_r\varepsilon_0\mathbf{E}.$$

Both signs flip when $\varepsilon_r, \mu_r < 0$. The triad $(\mathbf{E}, \mathbf{H}, \mathbf{k})$ becomes left-handed instead of right-handed. The Poynting vector $\mathbf{S} = \mathbf{E} \times \mathbf{H}$ now points *opposite* to $\mathbf{k}$.

For an outgoing wave with $\mathbf{S}$ pointing away from the source, $\mathbf{k}$ points toward the source. The refractive index $n_{\text{eff}} = c|\mathbf{k}|/\omega$ is conventionally signed by the direction of $\mathbf{k}$ relative to the outward normal. With $\mathbf{k}$ inward and $\mathbf{S}$ outward, this is precisely the negative-index branch:

$$n_{\text{eff}}(\omega) = -\sqrt{\varepsilon_r(\omega)\mu_r(\omega)} \quad \text{when } \varepsilon_r(\omega), \mu_r(\omega) < 0.$$

This is the Pendry 2000 negative refractive index. The choice is FORCED by causal outgoing-wave selection given the doubly-negative constitutive response; no separate postulate is required.

4.8 Substrate-level meaning of negative index

The substrate's rule-type structure in the doubly-negative band is engineered such that channels propagate with phase advancing in the direction *opposite* to energy flow. In substrate terms: the engineered combination of collective rule-type oscillation (wire-array Drude response) and resonant rule-type circulation (SRR Lorentz response) produces a coarse-grained substrate whose channels carry phase upstream relative to their energy flow. Negative refraction at an interface — bending to the same side of the normal as the incident wave — is the direct geometric consequence at the interface boundary.

4.9 Bandwidth and dispersion bounds

The wire-array and SRR responses are both intrinsically dispersive: by Kramers-Kronig relations on causal linear response, $\text{Im } \varepsilon(\omega) \neq 0$ over any band of finite width implies $\text{Re } \varepsilon(\omega)$ varies with ω . The doubly-negative band, where both $\text{Re } \varepsilon$ and $\text{Re } \mu$ are negative, is intrinsically narrow ($\Delta\omega/\omega \sim 5\text{--}20\%$ in typical microwave metamaterials). Lossless negative-index media over arbitrarily broad bandwidth are forbidden by causality.

5 Substrate Deformation, Effective Metric, and the Pendry Cloak

5.1 Setup

Under P-MM-1, P-MM-3, P-MM-4, P-MM-6, consider a slow substrate deformation $x'^i = f^i(x)$ with Jacobian

$$J^i_j = \frac{\partial x'^i}{\partial x^j}, \quad (J^{-1})^k_l = \frac{\partial x^k}{\partial x'^l}, \quad \det J > 0.$$

The undeformed substrate carries an isotropic vacuum rule-type response $\varepsilon^{ij}(x) = \varepsilon_0 \delta^{ij}$, $\mu^{ij}(x) = \mu_0 \delta^{ij}$. The deformation maps this to a coarse-grained anisotropic medium with transformed constitutive tensors derived below.

5.2 Rule-type deformation tensor

The substrate's local rule-type orientation is a rank-2 contravariant tensor structure $D^{ij}(x)$ at each point. Under deformation, D^{ij} transports covariantly as

$$D'^{ij}(x') = \frac{1}{\det J} J^i_k J^j_l D^{kl}(x).$$

The factor $1/\det J$ accounts for the volume change under deformation (density of weight one). The Jacobian factors transport the tensor indices into the deformed frame.

Conjugate tensor. The conjugate rank-2 covariant tensor structure $D_{ij}(x)$ transforms as

$$D'_{ij}(x') = (\det J) (J^{-1})^k_i (J^{-1})^l_j D_{kl}(x).$$

5.3 Constitutive transformation

The substrate's coarse-grained electric and magnetic responses are rank-2 contravariant tensors. Applying the deformation rule of §5.2:

$$\boxed{\varepsilon'^{ij}(x') = \frac{1}{\det J} J^i_k J^j_l \varepsilon^{kl}(x), \quad \mu'^{ij}(x') = \frac{1}{\det J} J^i_k J^j_l \mu^{kl}(x).}$$

This is the **transformation-optics constitutive law**. It is a substrate-level FORCED statement: given the deformation f , the transformed constitutive tensors are computed by applying the Jacobian densities. No separate posit is required.

Special case: vacuum starting medium. If the undeformed substrate is vacuum ($\varepsilon^{ij} = \varepsilon_0 \delta^{ij}$, $\mu^{ij} = \mu_0 \delta^{ij}$), then

$$\varepsilon'^{ij}(x') = \frac{\varepsilon_0}{\det J} (J J^T)^{ij}, \quad \mu'^{ij}(x') = \frac{\mu_0}{\det J} (J J^T)^{ij}.$$

The impedance $\sqrt{\mu'/\varepsilon'}$ equals $\sqrt{\mu_0/\varepsilon_0} = Z_0$ everywhere — the transformed medium is **impedance-matched to vacuum at every point**, eliminating reflections from interior surfaces.

5.4 Effective metric

The wave equation in the deformed substrate, with the transformed constitutive tensors, can be rewritten in covariant Laplace-Beltrami form. Consider Maxwell's equations in a transformed medium; using the transformation rule of §5.3 in the scalar-form wave equation gives

$$\partial_{x'^i} \left[\frac{(JJ^T)^{ij}}{\det J} \partial_{x'^j} \psi \right] + k_0^2 \psi = 0.$$

Identify

$$g^{ij}(x') = \frac{(JJ^T)^{ij}}{\det J}, \quad g_{ij}(x') = (J^{-1})^k{}_i (J^{-1})^l{}_j \delta_{kl}, \quad \sqrt{|g|} = \frac{1}{\det J}.$$

The wave equation becomes

$$\frac{1}{\sqrt{|g|}} \partial_{x'^i} \left[\sqrt{|g|} g^{ij} \partial_{x'^j} \psi \right] + k_0^2 \psi = 0.$$

This is the **covariant Laplace-Beltrami wave equation** on a Riemannian manifold with metric g_{ij} .

Crucial scope note. The metric g_{ij} is a coarse-grained rule-type metric on the substrate: it encodes the coarse-grained rule-type geometry seen by the propagating field. It is *not* spacetime curvature. The substrate-level operation that produced it is an engineered displacement (P-MM-3), not a gravitational effect. The identification $g_{ij} \leftrightarrow$ rule-type metric is structural, not analogical: the same mathematical object appears in both contexts because both contexts describe wave propagation through a curved geometry, but the underlying physics is different.

5.5 Eikonal limit: rays as geodesics

In the eikonal limit $\lambda \rightarrow 0$, the wave equation reduces to the eikonal equation

$$g^{ij} \partial_i S \partial_j S = n^2,$$

where $S(x')$ is the phase function and n is the local refractive index. The solutions are geodesics of the effective metric g_{ij} . **Light in a transformation-optics medium follows geodesics of the engineered substrate metric.** This is the operational statement that drives cloaking, lensing, and waveguiding in the transformation-optics regime.

5.6 Pendry spherical cloak: deformation

Choose the radial map

$$r' = R_1 + \alpha r, \quad \alpha = \frac{R_2 - R_1}{R_2}, \quad 0 \leq r \leq R_2,$$

in spherical coordinates (r, θ, ϕ) , with angular coordinates unchanged. The map sends the open ball $r < R_2$ onto the shell $R_1 < r' < R_2$. The interior point $r = 0$ maps to the inner shell $r' = R_1$; the outer boundary $r = R_2$ maps to itself.

The Jacobian in spherical coordinates has principal components:

- Radial: $\partial r' / \partial r = \alpha$.
- Polar angular: r' / r .
- Azimuthal angular: $(r' / r) \sin \theta / \sin \theta = r' / r$.

Note $r = (r' - R_1) / \alpha$. So $r' / r = \alpha r' / (r' - R_1)$.

5.7 Pendry cloak: constitutive tensors

Applying the constitutive transformation of §5.3 in spherical coordinates with $\det J = \alpha(r'/r)^2 = \alpha(\alpha r'/(r' - R_1))^2$:

$$\det J = \alpha^3 \frac{r'^2}{(r' - R_1)^2}.$$

The radial constitutive component:

$$\frac{\varepsilon_{\hat{r}'}}{\varepsilon_0} = \frac{J_{rr}^2}{\det J} \cdot \frac{1}{(g_{\text{sph}})_{rr}} \Rightarrow \frac{\varepsilon_{\hat{r}'}}{\varepsilon_0} = \beta \frac{(r' - R_1)^2}{r'^2},$$

where $\beta = R_2/(R_2 - R_1) = 1/\alpha$ (with appropriate handling of the spherical metric factors).

The angular constitutive components:

$$\frac{\varepsilon_{\hat{\theta}'}}{\varepsilon_0} = \frac{\varepsilon_{\hat{\phi}'}}{\varepsilon_0} = \beta.$$

The same expressions hold for μ/μ_0 — the cloak is impedance-matched to vacuum at every point. In boxed form:

$$\boxed{\frac{\varepsilon_{\hat{r}'}}{\varepsilon_0} = \beta \frac{(r' - R_1)^2}{r'^2}, \quad \frac{\varepsilon_{\hat{\theta}'}}{\varepsilon_0} = \frac{\varepsilon_{\hat{\phi}'}}{\varepsilon_0} = \beta, \quad \beta = \frac{R_2}{R_2 - R_1}.}$$

These are the **Pendry spherical cloak constitutive tensors** (Pendry 2006).

5.8 Effective metric of the cloak

The effective metric components, applying §5.4 to the cloak deformation:

$$g_{\hat{r}'\hat{r}'} = \beta^2 \frac{(r' - R_1)^2}{r'^2}, \quad g_{\hat{\theta}'\hat{\theta}'} = g_{\hat{\phi}'\hat{\phi}'} = \beta^2.$$

At $r' = R_1$, the radial metric component vanishes: $g_{\hat{r}'\hat{r}'}(r' = R_1) = 0$.

5.9 Substrate-level meaning of invisibility

The vanishing of $g_{\hat{r}'\hat{r}'}$ at $r' = R_1$ has the substrate-level reading that **rule-type channels carrying field amplitude cannot enter the cloaked interior**. The radial distance, in the rule-type metric, becomes singular at the inner shell: any geodesic of g_{ij} that attempts to cross $r' = R_1$ would require infinite proper length. The interior is *topologically excluded* from the substrate seen by the propagating field.

This is the structural meaning of invisibility:

- Not absorption (no energy loss in the cloak shell).
- Not reflection (impedance matching eliminates reflections).
- Not concealment (no hiding behind material screens).
- **Topological exclusion:** the cloaked region is removed from the rule-type geometry seen by the propagating channel.

Energy flow follows geodesics of g_{ij} , which detour smoothly around the cloaked interior and exit the outer shell $r' = R_2$ as if traveling through vacuum. An observer outside the cloak sees the field profile of vacuum propagation; the cloaked object is invisible.

5.10 General transformation optics

The Pendry cloak is one instance of the general transformation-optics construction. Any smooth deformation $f : x \mapsto x'$ with J and J^{-1} bounded and $\det J > 0$ produces a substrate-level effective medium with constitutive tensors given by §5.3 and effective metric given by §5.4. The geodesics of the effective metric are the engineered light rays.

Physical realizability conditions:

- Smoothness of f : J, J^{-1} smooth and bounded.
- Orientation preservation: $\det J > 0$.
- Homogenization regime: the resulting ε', μ' must be achievable by some periodic microstructure with $a \ll \lambda$ at the operating frequency.
- Absence of unintentional singularities: ε' and μ' singularities only at designed locations.
- Bandwidth: dispersion of the implementing microstructure restricts the design to a finite frequency band.

The Pendry cloak saturates condition 4 deliberately at $r' = R_1$; this singularity is the price of perfect cloaking, and it is what makes the perfect cloak intrinsically bandwidth-limited and physically realizable only as an approximation.

6 Metasurfaces and the Generalized Snell's Law

6.1 Codimension-1 discontinuity

Under P-MM-5, P-MM-6, consider a surface Σ at $z = 0$ across which the substrate's rule-type structure changes discontinuously. The bulk regions $z < 0$ and $z > 0$ are described by the coarse-grained wave equation of P-MM-4 with refractive indices n_1 and n_2 respectively. The discontinuity at Σ imposes jump conditions on the wave field.

6.2 Passive interface: standard Snell's law

For a passive interface (no engineered surface phase profile, no surface polarization, no surface currents), the standard tangential-field continuity holds:

$$\mathbf{E}_{\parallel}(r_{\parallel}, 0^+) = \mathbf{E}_{\parallel}(r_{\parallel}, 0^-), \quad \mathbf{H}_{\parallel}(r_{\parallel}, 0^+) = \mathbf{H}_{\parallel}(r_{\parallel}, 0^-).$$

For a scalar wave ψ (component of $\mathbf{E}$ or $\mathbf{H}$ parallel to Σ):

$$\psi(r_{\parallel}, 0^+) = \psi(r_{\parallel}, 0^-).$$

Take a plane wave on side 1: $\psi_1 = e^{ik_{\parallel,1}r_{\parallel} + ik_{z,1}z}$. Continuity at $z = 0$ forces the tangential wavevector to match: $k_{\parallel,2} = k_{\parallel,1}$. With $k_{\parallel,1} = n_1 k_0 \sin \theta_i$ and $k_{\parallel,2} = n_2 k_0 \sin \theta_t$, the matching condition gives

$$n_1 \sin \theta_i = n_2 \sin \theta_t,$$

the standard Snell's law.

6.3 Engineered phase profile

A metasurface is a codimension-1 surface engineered to imprint a tangential phase profile $\Phi(r_{\parallel})$ on the transmitted field. The phase profile is produced by subwavelength resonant scatterers (“meta-atoms”) whose engineered geometry imprints a position-dependent phase shift. Each meta-atom is a coarse-grained rule-type discontinuity of negligible thickness; the metasurface as a whole is the spatial pattern of these meta-atoms.

The boundary condition at Σ becomes, for a scalar wave:

$$\boxed{\psi_2(r_{\parallel}, 0^+) = \psi_1(r_{\parallel}, 0^-) e^{i\Phi(r_{\parallel})}.$$

Substrate-level meaning. The metasurface is a codimension-1 engineered rule-type discontinuity. The tangential phase profile $\Phi(r_{\parallel})$ is the integrated rule-type phase imprinted on a channel as it crosses Σ . The function Φ is engineered by the meta-atom geometry: each meta-atom resonance produces a controlled phase shift, and the spatial arrangement of meta-atoms produces the spatial profile.

6.4 Tangential momentum kick

Take an incident plane wave on side 1: $\psi_1 = e^{ik_{\parallel,1}r_{\parallel} + ik_{z,1}z}$. At $z = 0^-$, the field is $\psi_1(r_{\parallel}, 0^-) = e^{ik_{\parallel,1}r_{\parallel}}$. Applying the metasurface boundary condition:

$$\psi_2(r_{\parallel}, 0^+) = e^{ik_{\parallel,1}r_{\parallel} + i\Phi(r_{\parallel})}.$$

Take the tangential gradient of the phase. For a *linear* phase profile $\Phi(x) = (d\Phi/dx)x$:

$$\psi_2(r_{\parallel}, 0^+) = e^{i[k_{\parallel,1} + d\Phi/dx]x},$$

showing that the tangential wavevector on side 2 is

$$\boxed{k_{\parallel,2} = k_{\parallel,1} + \nabla_{\parallel}\Phi(r_{\parallel}).$$

The metasurface imparts a **tangential momentum kick** equal to the gradient of the engineered phase profile. This is the operational signature of the metasurface: the discontinuity adds a tangential wavevector contribution that no passive interface can produce.

6.5 Generalized Snell's law

Substitute $k_{\parallel,1} = n_1 k_0 \sin \theta_i$, $k_{\parallel,2} = n_2 k_0 \sin \theta_t$, $k_0 = 2\pi/\lambda$ into the matching condition $k_{\parallel,2} = k_{\parallel,1} + d\Phi/dx$:

$$n_2 k_0 \sin \theta_t = n_1 k_0 \sin \theta_i + \frac{d\Phi}{dx},$$

or equivalently

$$\boxed{n_1 \sin \theta_i = n_2 \sin \theta_t + \frac{\lambda}{2\pi} \frac{d\Phi}{dx}.$$

This is the **Capasso 2011 generalized Snell's law**. The standard Snell's law is the special case $d\Phi/dx = 0$.

6.6 Consequences

The generalized Snell's law permits:

- **Anomalous refraction.** For $d\Phi/dx \neq 0$, the refraction angle θ_t is no longer determined solely by the bulk indices and incidence angle. Refraction angles unattainable from n_1, n_2 alone become accessible.
- **Negative refraction at a single interface.** A sufficiently steep phase gradient $d\Phi/dx$ can produce θ_t on the same side of the normal as θ_i even with both bulk indices positive.
- **Reflectionless beam steering.** By engineering Φ , the metasurface redirects a transmitted beam to any angle within the limits of the phase gradient.
- **Phase-only optical components.** Lenses, waveplates, holograms, and arbitrary optical functionality implemented as flat metasurfaces of subwavelength thickness.

6.7 Limits

The phase profile $\Phi(r_{\parallel})$ must vary slowly on the scale of λ for the boundary-condition treatment to be valid. Phase profiles with features finer than λ are not captured by the metasurface picture alone; they enter full diffraction theory. The phase imprint is intrinsically frequency-dependent (the meta-atoms supporting Φ are resonant), so achromatic metasurfaces require multi-resonance engineering and remain bandwidth-limited.

7 Photonic Bandgaps via Bloch Eigenmodes

7.1 When homogenization breaks down

The two-scale expansion of §3 requires $a/\lambda \rightarrow 0$. At $a \sim \lambda$, the expansion no longer converges term-by-term: the corrector field develops slow-coordinate dependence, and additional corrections of order $(a/\lambda)^2$ enter the effective coefficient. The relevant description in this regime is Bloch / Floquet analysis: instead of effective constants, one obtains a band structure $\omega_n(k)$ on the Brillouin zone.

This is the regime of **photonic crystals**, adjacent to the strict homogenization regime.

7.2 Bloch's theorem on the periodic substrate

Under P-MM-1, P-MM-2, P-MM-4, P-MM-6 with $a \sim \lambda$, consider the wave equation

$$\partial_i [A^{ij}(x) \partial_j \psi] + \omega^2 n^2(x) \psi = 0,$$

with A^{ij} and n^2 periodic with lattice $\Lambda = a\mathbb{Z}^3$. Define the translation operator T_R acting on functions: $T_R \psi(x) = \psi(x + R)$ for $R \in \Lambda$.

The wave operator commutes with T_R for any $R \in \Lambda$. The simultaneous eigenfunctions of the wave operator and $\{T_R\}_{R \in \Lambda}$ form a complete basis. The translation operators are unitary on a suitable Hilbert space, so their eigenvalues are phases $e^{ik \cdot R}$ with k a real vector. Restricting k to the first Brillouin zone (BZ) — the set of k such that $|k| \leq |k - G|$ for all reciprocal lattice vectors $G \in \Lambda^*$ — uniquely labels the simultaneous eigenfunctions.

Bloch form. Simultaneous eigenfunctions take the form

$$\boxed{\psi_{n,k}(x) = u_{n,k}(x) e^{ik \cdot x}, \quad u_{n,k}(x + R) = u_{n,k}(x) \text{ for all } R \in \Lambda,}$$

indexed by band index $n \in \mathbb{N}$ and crystal momentum k in the first BZ.

7.3 Band structure

Substituting the Bloch form into the wave equation gives, after expanding the derivatives,

$$(\partial_i + ik_i) [A^{ij}(x) (\partial_j + ik_j) u_{n,k}] + \omega_n^2(k) n^2(x) u_{n,k} = 0,$$

a Hermitian generalized eigenvalue problem for $u_{n,k}$ on a single unit cell with periodic boundary conditions. At each k , the eigenvalues $\omega_n^2(k)$ form a discrete real sequence

$$0 \leq \omega_1^2(k) \leq \omega_2^2(k) \leq \omega_3^2(k) \leq \dots,$$

indexed by band number n . As k varies continuously over the BZ, each eigenvalue $\omega_n^2(k)$ traces out the n -th **band** of the band structure.

7.4 Bandgap formation

A **bandgap** is a frequency interval $[\omega_a, \omega_b]$ such that no value of k produces an eigenvalue $\omega_n^2(k)$ in $[\omega_a^2, \omega_b^2]$ for any band n . At those frequencies, no propagating Bloch eigenmode exists — the substrate forbids field propagation in the bulk.

Mechanism. A bandgap appears when the bands $\omega_n(k)$ and $\omega_{n+1}(k)$, in some region of the BZ, separate without crossing. The standard mechanism is constructive vs. destructive interference of the field across the unit cell: at certain k near the BZ boundary, the spatial pattern of the lower band has nodes on the high-index regions while the higher band has nodes on the low-index regions, with the energy difference producing a gap.

For periodic dielectric substrates with sufficient index contrast (typically $\Delta\epsilon/\epsilon \gtrsim 0.5$ for full 3D gaps), complete bandgaps appear: frequency intervals in which no propagating mode exists in any direction. This is the **Yablonovitch photonic bandgap**.

7.5 Substrate-level meaning

The bandgap is a frequency window in which the periodic rule-type substrate offers no allowed coarse-grained channel. The substrate's rule-type structure, periodicized at scale $a \sim \lambda$, supports certain frequencies as propagating modes (the bands) and forbids others (the gaps). The forbidden frequencies are those for which no Bloch eigenmode can be assembled from the local rule-type response.

This is the substrate-level analog of an electronic bandgap in a crystalline solid: same Bloch machinery, applied to the substrate's coarse-grained rule-type response rather than electron wave functions.

7.6 Relation to homogenization

The homogenization regime $a \ll \lambda$ corresponds to the long-wavelength limit $k \rightarrow 0$ of the lowest band:

$$\omega_1(k) \approx c_{\text{eff}}|k|, \quad k \rightarrow 0,$$

with c_{eff} determined by the effective constitutive tensors of §§3-4. The lowest band, near $k = 0$, looks like vacuum propagation in a homogeneous effective medium.

Bandgaps live further into the BZ, at $|k| \sim \pi/a$, where homogenization is no longer valid and the full Bloch structure is needed. At these wavevectors, the wavelength $\lambda = 2\pi/k \sim 2a$ is comparable to the lattice period, and the field oscillates significantly within one cell.

The two regimes — homogenization and Bloch / bandgap — are not independent. They are two limits of the same substrate operation: engineered periodic microstructure under P-MM-1, P-MM-2. Homogenization gives the long-wavelength effective medium; Bloch analysis gives the full band structure including bandgaps. A photonic crystal is the same substrate operation as a homogenized metamaterial; the difference is the operating wavelength relative to the lattice period.

7.7 Defects and waveguiding

Local defects in the periodic substrate (a single perturbed unit cell, a missing inclusion, an embedded line of perturbed cells) support localized modes inside the bandgap. These defect states are the substrate-level mechanism for photonic-crystal waveguides, cavities, and add-drop filters. The defect modifies the local rule-type response, producing a discrete eigenmode whose frequency lies in the bulk's forbidden gap.

8 The Unifying Substrate-Level Statement

8.1 Three control axes

The three closures derived in this walkthrough — homogenization, transformation optics, metasurface boundary conditions — correspond to three independent engineered modulations of the substrate’s coarse-grained rule-type structure:

Operation	Substrate primitive	Coarse-grained result	Nobel-frontier instance
Smooth periodic microstructure	P-MM-1, P-MM-2	$\varepsilon_{\text{eff}}, \mu_{\text{eff}}$ (and Bloch bands at $a \sim \lambda$)	Pendry-negative-index, Yablonovitch-bandgap
Smooth deformation	P-MM-3	effective metric g_{ij} , transformed ε, μ	Pendry-cloak
Codim-1 discontinuity	P-MM-5	tangential momentum kick $\nabla_{\parallel} \Phi$	Capasso-metasurface

8.2 Exhaustiveness within the precursor machinery

Within the coarse-graining window P-MM-6 and under the second-order wave equation P-MM-4, engineered modifications of the substrate enter one of three locations:

1. The coefficient $A^{ij}(x)$ varies smoothly on a scale a — Closure 1.
2. The coordinate x is the image of a deformed substrate $x = f(X')$ with f smooth — Closure 2.
3. The coefficient $A^{ij}(x)$ has a codimension-1 jump on a surface — Closure 3.

Higher-codimension features (lines, points) do not survive coarse-graining: they appear as point scatterers at coarse-grained scales and are not metamaterial control mechanisms. Volumetric non-smooth features cannot be averaged within the homogenization window and reduce to one of the three cases at the operating wavelength. **The three closures therefore exhaust the substrate-level wave-control space within the precursor machinery.**

8.3 The unifying identity

Metamaterials = engineered rule-type microstructure + engineered deformation + engineered discontinuity.

Each Nobel-frontier result is the instantiation of one axis in isolation:

- **Yablonovitch bandgap** = periodic microstructure at $a \sim \lambda$.
- **Pendry negative index** = doubly-resonant periodic microstructure at $a \ll \lambda$.
- **Pendry cloak** = smooth deformation with FORCED constitutive transformation.
- **Capasso metasurface** = engineered codim-1 phase discontinuity.

Practical metamaterials generally compose all three axes: a transformation-optics device implemented by a graded homogenized medium with a metasurface termination, for example.

8.4 The unified substrate-level wave equation

The most general coarse-grained wave equation in the precursor machinery is

$$\frac{1}{\sqrt{|g|}} \partial_i \left[\sqrt{|g|} g^{ij} (\varepsilon_{\text{eff}}^{-1})^{jk} \partial_k \psi \right] + k_0^2 \mu_{\text{eff}} \psi = 0,$$

subject to jump conditions on engineered codim-1 surfaces Σ :

$$\psi(r_{\parallel}, 0^+) = \psi(r_{\parallel}, 0^-) e^{i\Phi(r_{\parallel})}.$$

This is the **unified substrate-level wave equation of the Arc**. The three closures are its three independent control knobs: $\varepsilon_{\text{eff}}, \mu_{\text{eff}}$ (Closure 1), g_{ij} (Closure 2), and Φ on Σ (Closure 3). All three knobs derive from the same substrate operations (P-MM-1, P-MM-3, P-MM-5) coarse-grained under P-MM-6 and propagated under P-MM-4.

8.5 Scope and limits

The closures hold under the following bounds:

- **Homogenization regime:** $a/\lambda \ll 1$. Beyond $a \sim \lambda/3$ the leading expansion is qualitative only; Bloch analysis takes over.
- **Negative-index bandwidth:** $\varepsilon_r < 0$ and $\mu_r < 0$ overlap over a band intrinsically narrowed by Kramers-Kronig. Lossless negative-index media over arbitrary bandwidth are forbidden.
- **Transformation optics smoothness:** J and J^{-1} bounded, $\det J > 0$. Singular maps (Pendry cloak at $r' = R_1$) are intrinsically dispersive and bandwidth-limited.
- **Metasurface phase variation:** $\Phi(r_{\parallel})$ slow on λ . Sub- λ phase features require full diffraction theory.
- **Universal:** energy conservation, causality (Kramers-Kronig), Sommerfeld-Brillouin signal-front bound $v_{\text{front}} \leq c$, bandwidth-response trade-off across all axes.

The effective metric g_{ij} of §5 is a coarse-grained rule-type metric on the substrate — not spacetime curvature. The substrate-level operation that produces it is engineered displacement, not gravitational physics. The identification is structural, not analogical.

9 FORCED / INHERITED / OPEN Accounting

9.1 FORCED at substrate level

The following results are FORCED by the substrate primitives P-MM-1 through P-MM-6 and the derivations of this walkthrough. No additional posit is required.

- Two-scale lift and derivative splitting (§3.2) — FORCED from P-MM-2, P-MM-6.
- Order-by-order equations (§3.3) — FORCED.
- Leading-order $\psi_0(X, y) = \psi_0(X)$ independence from fast variable (§3.3) — FORCED by positive-definiteness of A^{ij} .
- Cell problem with Y -periodicity and zero-mean corrector (§3.3) — FORCED.
- Cell-averaging operator properties (vanishing of pure y -divergences, commutativity with X -derivatives) (§3.4) — FORCED.
- Effective-coefficient formula $A^{*ik} = \langle A^{ik} \rangle + \langle A^{ij} \partial_{y^j} \chi^k \rangle$ (§3.5) — FORCED.
- Voigt-Reuss bounds (§3.6) — FORCED.
- Anisotropy from microstructure (§3.7, §4.2) — FORCED FORM.
- Form of effective constitutive tensors $\varepsilon_{\text{eff}}^{ij}, \mu_{\text{eff}}^{ij}$ (§4.2) — FORCED FORM.
- Negative-index branch choice $n = -\sqrt{\varepsilon_r \mu_r}$ in the doubly-negative band (§4.7) — FORCED by causal outgoing-wave selection.
- Constitutive transformation under deformation $\varepsilon'^{ij} = (\det J)^{-1} J^i_k J^j_l \varepsilon^{kl}$ (§5.3) — FORCED from P-MM-3 + P-MM-4 + P-MM-6.
- Impedance matching to vacuum under deformation from vacuum starting medium (§5.3) — FORCED.
- Effective metric $g_{ij} = (J^{-1})^k_i (J^{-1})^l_j \delta_{kl}$ (§5.4) — FORCED.
- Light rays as geodesics of g_{ij} in eikonal limit (§5.5) — FORCED.
- Pendry cloak constitutive tensors (§5.7) — FORCED FORM from the deformation $r' = R_1 + \alpha r$.
- Topological exclusion of cloaked interior (§5.9) — FORCED by vanishing of $g_{\hat{r}'\hat{r}'}$ at $r' = R_1$.

- Metasurface phase-imprinting boundary condition $\psi_2 = \psi_1 e^{i\Phi}$ (§6.3) — FORCED from P-MM-5 + P-MM-6.
- Tangential momentum kick $k_{\parallel,2} = k_{\parallel,1} + \nabla_{\parallel}\Phi$ (§6.4) — FORCED.
- Generalized Snell's law (§6.5) — FORCED.
- Bloch form for periodic substrates (§7.2) — FORCED from P-MM-1, P-MM-2, P-MM-4.
- Band structure as Hermitian eigenvalue problem on the unit cell (§7.3) — FORCED.
- Photonic-bandgap mechanism as eigenvalue gaps in the Bloch problem (§7.4) — FORCED FORM.
- Continuity of homogenization regime with Bloch regime through long-wavelength limit of lowest band (§7.6) — FORCED.
- Three-axis exhaustiveness within P-MM-4 + P-MM-6 (§8.2) — FORCED.
- Unifying identity (§8.3) — FORCED.
- Unified substrate-level wave equation (§8.4) — FORCED FORM.

9.2 INHERITED (form FORCED, values from microstructure or geometry choice)

The following take their *form* from the FORCED results above, but their numerical values are INHERITED from specific microstructure or device-geometry choices made by the engineer.

- Drude plasma frequency $\omega_p^2 = 2\pi c^2/[a^2 \ln(a/r_0)]$ (§4.4) — Drude FORM FORCED; coefficient INHERITED from wire-array geometry.
- Lorentz parameters F, ω_0, γ (§4.5) — Lorentz FORM FORCED; oscillator parameters INHERITED from SRR geometry.
- Cloak parameters α, β (§5.7) — INHERITED from cloak-shell radii R_1, R_2 .
- Metasurface phase profile $\Phi(r_{\parallel})$ (§6.3) — INHERITED from meta-atom geometry and arrangement.
- Band structure $\omega_n(k)$ (§7.3) — FORM FORCED; specific band positions INHERITED from periodic geometry and material contrast.
- Bandgap widths and locations (§7.4) — INHERITED from periodic geometry.
- Corrector fields $\chi^j(y)$ (§3.3) — equation FORCED; specific solutions INHERITED from cell coefficient $A^{ij}(y)$.

9.3 OPEN

The following items are OPEN at the substrate level. They are not derived in this walkthrough and are flagged for future work.

- Higher-order homogenization corrections at $a \sim \lambda/3$ to λ . The leading expansion is FORCED; next-order corrections at $(a/\lambda)^2$ are FORM-FORCED-VALUES-INHERITED from specific cell solutions but not derived here.
- Nonlocal / spatially dispersive metamaterials. The closure machinery is local; spatial dispersion enters at the next order in a/λ and requires extension beyond two-scale expansion.
- Active and time-varying metamaterials. Time-dependent deformation $u(X, t)$ and time-varying microstructure are allowed by P-MM-1 and P-MM-3 respectively, but the corresponding closures (Floquet-in-time, parametric amplification, non-reciprocal phenomena) require their own derivations.
- Topological photonics within Closure 1. Periodic substrates with engineered band topology, Berry curvature on the BZ, Chern numbers, photonic topological insulators — adjacent to Closure 1 but not derived here.
- Quantum-optical metamaterials. Single-photon and entangled-photon propagation through metamaterials uses the same coarse-grained wave equation at the field level; substrate-level quantum-state evolution through Closure 1/2/3 devices is reachable but not derived here.

- Bulk-boundary correspondence for photonic systems. The relation between bulk band topology and boundary states is not derived here.
- Hyperbolic metamaterials. Substrates with effective tensors of mixed-signature signature ($\varepsilon_{\parallel} > 0$, $\varepsilon_{\perp} < 0$ or vice versa) are within Closure 1's reach but not derived in detail here.
- Time-domain metasurfaces. Spatiotemporal phase profiles $\Phi(r_{\parallel}, t)$ produce frequency shifts in addition to angle shifts; the static derivation of §6 does not cover this case.

No new substrate primitives are introduced anywhere in the walkthrough.

10 Exact Claims

The substrate-level claims established in this walkthrough are:

1. The substrate ontology FORCES a coarse-grained second-order wave equation (P-MM-4) whose coefficients are determined by the substrate's local rule-type content (P-MM-1).
2. Subwavelength periodic microstructure (P-MM-2) admits two-scale homogenization. The cell-problem corrector field $\chi^j(y)$, defined by an elliptic boundary-value problem on the unit cell with periodic boundary conditions and zero mean, produces effective constitutive tensors

$$\varepsilon_{\text{eff}}^{ij} = \langle \varepsilon^{ij} \rangle + \langle \varepsilon^{ik} \partial_{y^k} \chi_E^j \rangle, \quad (\mu_{\text{eff}}^{-1})^{ij} = \langle (\mu^{-1})^{ij} \rangle + \langle (\mu^{-1})^{ik} \partial_{y^k} \chi_M^j \rangle.$$
3. The effective tensors satisfy Voigt-Reuss bounds. Microstructure produces anisotropy from isotropic constituents.
4. Engineered wire-array microstructures produce a Drude effective dielectric function with plasma frequency $\omega_p^2 = 2\pi c^2 / [a^2 \ln(a/r_0)]$, giving $\varepsilon_{\text{eff}}(\omega) < 0$ for $\omega < \omega_p$.
5. Engineered split-ring resonator microstructures produce a Lorentz effective magnetic function with $\mu_{\text{eff}}(\omega) < 0$ in a band immediately above the resonance frequency ω_0 .
6. In the doubly-negative band, the negative refractive index $n_{\text{eff}} = -\sqrt{\varepsilon_r \mu_r}$ is FORCED by causal outgoing-wave selection. No separate postulate is required.
7. Slow substrate deformation (P-MM-3) produces transformed constitutive tensors $\varepsilon'^{ij} = (\det J)^{-1} J^i_k J^j_l \varepsilon^{kl}$ and an effective metric $g_{ij} = (J^{-1})^k_i (J^{-1})^l_j \delta_{kl}$ on the wave equation, written in covariant Laplace-Beltrami form.
8. The Pendry spherical cloak deformation $r' = R_1 + \alpha r$ with $\alpha = (R_2 - R_1)/R_2$ produces constitutive tensors $\varepsilon_{\hat{r}'} / \varepsilon_0 = \beta(r' - R_1)^2 / r'^2$, $\varepsilon_{\hat{\theta}'} / \varepsilon_0 = \varepsilon_{\hat{\phi}'} / \varepsilon_0 = \beta$. The cloaked interior is topologically excluded at substrate level by the vanishing of $g_{\hat{r}'\hat{r}'}$ at $r' = R_1$.
9. The effective metric g_{ij} is a coarse-grained rule-type metric on the substrate. It is *not* spacetime curvature; no claim about gravitational physics is made or required.
10. Engineered codim-1 phase discontinuities (P-MM-5) imprint a tangential momentum kick $k_{\parallel,2} = k_{\parallel,1} + \nabla_{\parallel} \Phi$ on transmitted channels. The generalized Snell's law $n_1 \sin \theta_i = n_2 \sin \theta_t + (\lambda/2\pi)(d\Phi/dx)$ follows directly.
11. Periodic substrate at $a \sim \lambda$ supports Bloch eigenmodes $\psi_{n,k} = u_{n,k} e^{ik \cdot x}$ with band structure $\omega_n(k)$ on the Brillouin zone. Photonic bandgaps are frequency intervals where no eigenvalue exists, FORCED-FORM at substrate level.
12. The three substrate-level wave-control axes (microstructure, deformation, discontinuity) exhaust the precursor-machinery design space within P-MM-6.
13. Practical metamaterials are engineered compositions along the three axes. The unified substrate-level wave equation is

$$\frac{1}{\sqrt{|g|}} \partial_i \left[\sqrt{|g|} g^{ij} (\varepsilon_{\text{eff}}^{-1})^{jk} \partial_k \psi \right] + k_0^2 \mu_{\text{eff}} \psi = 0,$$

with jump conditions $\psi(0^+) = \psi(0^-) e^{i\Phi(r_{\parallel})}$ on engineered codim-1 surfaces.

14. The precursor machinery is bounded by the coarse-graining window P-MM-6, Kramers-Kronig dispersion-bandwidth trade-offs, and the Sommerfeld-Brillouin signal-front bound $v_{\text{front}} \leq c$.
15. The four Nobel-frontier metamaterials and photonics directions — Yablonovitch's photonic bandgaps, Pendry's negative refractive index, Pendry's electromagnetic cloak, and Capasso's generalized Snell's law — are reproduced at the substrate level by the three precursor closures plus the adjacent Bloch / bandgap regime, with no new substrate primitives.

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